

# Xiaofeng Qiao

Ph.D. Student in Computer Science | AI Systems, Efficient ML, Reliable AI  
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## RESEARCH PROFILE

Ph.D. student and researcher focused on high-performance and reliable AI systems, including GPU runtime optimization, compiler and operator optimization, structured model compression, uncertainty-aware learning, and AI inference infrastructure.

## TECHNICAL SKILLS

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|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>            | Python, C++, C, Go, Java, Scala, Swift, SQL                                                                                                    |
| <b>AI Systems</b>             | CUDA, NVIDIA PTX, NVIDIA Nsight Profiler, TVM, TensorRT, Triton, vLLM, SGLang, Megatron                                                        |
| <b>ML / AI</b>                | PyTorch, TensorFlow, model pruning, mixed-precision training, post-training quantization, knowledge distillation, VLM, VLA, RAG, agent systems |
| <b>Infrastructure / Tools</b> | AWS, Docker, Kubernetes, Git, ROS, MCP, NumPy, Pandas, MySQL, Oracle SQL, MongoDB, Selenium, Wireshark                                         |

## EDUCATION

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| <b>George Mason University - Ph.D. in Computer Science</b><br>GPA: 3.78 / 4.00   Research focus: high-performance and secure AI systems | <b>Aug 2024 - May 2029</b><br>Fairfax, VA, USA  |
| <b>The Pennsylvania State University - M.S. in Computer Science</b><br>GPA: 3.95 / 4.00                                                 | <b>Aug 2021 - Jan 2024</b><br>Pennsylvania, USA |
| <b>University of Electronic Science and Technology of China - B.Eng. in Electronic Information Engineering</b><br>GPA: 3.92 / 4.00      | <b>Sep 2017 - Jun 2021</b><br>Chengdu, China    |

## RESEARCH EXPERIENCE

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| <b>GPU Runtime, High-Performance Computing, and Operator Optimization</b><br>George Mason University (Ph.D.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Aug 2024 - May 2029</b><br>Fairfax, VA, USA  |
| <ul style="list-style-type: none"><li>Designed and implemented heterogeneous-friendly AI infrastructure to optimize computation graphs and CUDA kernels, generating CUDA and inline PTX code.</li><li>Built TVM TIR templates for operators including Conv2D, matrix multiplication, attention, and max pooling, and connected optimized kernels to corresponding PyTorch functions.</li><li>Used XGBoost as a surrogate model and simulated annealing over operator-template search spaces, with diversity filtering to reduce local-optimum collapse; supported both single-objective and multi-objective Pareto optimization.</li><li>Built transformation infrastructure to lower optimized TIR operators to ONNX, including constant replacement, loop unrolling, and variable merging; implemented a C++ code-generation module that converts ONNX scripts to CUDA and inline PTX, compiled with NVCC.</li><li>Deployed optimized neural-network operators on H100 and RTX 4090 servers as well as Raspberry Pi and NVIDIA Jetson Nano edge devices; achieved about <b>60% lower</b> end-to-end latency than TensorFlow XLA and about <b>40% lower</b> latency than TensorRT in selected deployments.</li><li>Designed a software-defined GPU scheduling framework using inline PTX and persistent threads for warp-level scheduling, reducing end-to-end inference latency by about <b>38%</b>.</li><li>Extended TVM with operator fusion and software-defined GPU scheduling to automate optimization, reducing MobileNet-v3 end-to-end inference latency by about <b>32%</b>.</li><li>Implemented multithreaded and multiprocess concurrency plus power management for software-defined GPU scheduling. By controlling active threads, shared memory, and register allocation, extended stable peak-performance runtime on Raspberry Pi by <b>21%</b> versus DVFS-based methods while avoiding overheating.</li></ul> |                                                 |
| <b>Uncertainty-Aware Model Compression and Calibration</b><br>Research Assistant, The Pennsylvania State University                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Aug 2021 - Jan 2024</b><br>Pennsylvania, USA |
| <ul style="list-style-type: none"><li>Implemented a multi-scenario, uncertainty-aware model-compression framework using TensorFlow and TensorFlow-Slim; the work was supported by the U.S. National Science Foundation (NSF) and delivered as a sponsored research outcome.</li><li>Applied structured pruning across visual perception (ResNet-50, InceptionV3, SegNet), activity recognition (GRU, LSTM, Transformer), and control (A3C), with uncertainty estimated using Monte Carlo Dropout and deep ensembles.</li><li>Introduced mixed-precision training and post-training INT8 quantization, followed by knowledge distillation for post-compression fine-tuning. In human activity recognition, reduced model size by about <b>85%</b> with no more than a <b>2 percentage-point</b> accuracy drop.</li><li>For out-of-distribution and adversarial-sample detection, uncertainty calibration improved average AUROC by <b>12 percentage points</b> after compressing <b>66%</b> of model parameters.</li></ul> <p><b>Visual perception tasks</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                 |

- Designed an automated CNN backbone compression pipeline (e.g., ResNet-50) for visual classification, modifying and fine-tuning TensorFlow pretrained models. Pruning reduced model size to **55%** of the original without accuracy loss in selected experiments.
- Evaluated ResNet-50 under **16** visual corruptions such as Gaussian blur, motion blur, overexposure, and snow; uncertainty-corrected compressed models improved reliability metrics by **20 percentage points**.
- Used active learning, data augmentation, learning-rate adjustment, knowledge distillation, and soft labels to improve generalization, increasing accuracy on corrupted visual data by about **14 percentage points**.

#### **Activity recognition tasks**

- Implemented cross-modal human activity recognition using inertial measurement unit (IMU) signals and deployed the system to multiple smartwatches and smartphones.
- Mapped sensor streams into a unified space using Fourier transforms, concatenated features, and performed cross-modal fusion with GRU, LSTM, and Transformer modules. An adaptive pruning module combined with structured pruning, quantization, and distillation compressed **85%** of model parameters with less than a **1 percentage-point** accuracy drop.

#### **AI Applications and Inference Systems**

Recent

- Built and optimized a real-time GPU inference platform for the ChatTTSPlus TTS model. Combined structured pruning, quantization, knowledge distillation, and TVM-based graph optimization, operator fusion, loop unrolling, and memory-layout optimization to reduce end-to-end inference latency by **45%**.
- Used dynamic batching to improve throughput by **44%** and CUDA Streams to improve GPU utilization by **31%**.
- Integrated LLM services for conversational functionality and built AI infrastructure for input-length limits, rate limits, safe-response policies, context management (MCP), external-knowledge caching, and retrieval-augmented generation (RAG). These controls reduced token consumption by about **70%** while improving system safety, stability, and response controllability.